A COUNTERMODEL-FIRST STAGE-1 IMPLICATION SOLVER FOR MAGMA EQUATIONS

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Abstract. We study Stage-1 equational implication for magmas: given two equations E_1, E_2 over one binary operation, return a strict boolean answer to whether $E_1 \models E_2$. We formalize and evaluate a hybrid policy that combines exhaustive finite countermodel search (orders 1 through 3) with a bounded symbolic derivation module. The policy is countermodel-first: if a finite witness satisfies E_1 and refutes E_2, the output is `false`; otherwise the system attempts bounded derivation and outputs `true` when derivation succeeds; if neither branch resolves the query, the policy still returns `true` to satisfy the Stage-1 output contract.

Our first result is a structural guarantee: the implemented policy is total and always returns a value in $\{\texttt{true}, \texttt{false}\}$. Our second result is soundness of negative answers: every `false` output is accompanied by an explicit finite countermodel, hence is semantically correct. Our third result is computational: on a curated benchmark of ten labeled implication pairs, the method attains 10/10 accuracy, with Wilson 95% interval $[0.722, 1.000]$ and mean runtime 0.0623 seconds per query. All four negative instances are witnessed by order-2 magmas. These findings support a practical, reproducible baseline for fast implication triage prior to heavyweight theorem proving.

1. Introduction

Determining whether one equational law implies another is a basic operation in universal algebra and automated theorem proving. In the Equational Theories Project (ETP), this question appears at scale: a large implication graph over magma laws has been computed and formalized, making implication prediction and certification a central practical task [3]. Fast implication decisions are useful as a routing layer before expensive proof search.

Classical finite-model tools such as Mace4 emphasize falsification by small counterexamples [4]. In parallel, learning-based approaches report strong predictive performance for equational tasks [5, 1], and latent-geometry analyses reveal structure in the space of equational theories [2]. For Stage-1 settings, however, there remains a need for a transparent, reproducible baseline that always emits strict boolean answers while keeping runtime low.

We study single-premise implication queries over magmas: given equations E_1 and E_2, decide whether $E_1 \models E_2$ with output in $\{\texttt{true}, \texttt{false}\}$. Our solver uses a countermodel-first hybrid policy: bounded finite model search up to order 3, then bounded symbolic derivation, and finally a deterministic boolean fallback required by the task contract.

Our contributions are threefold.

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(1) We formalize the Stage-1 policy and prove structural guarantees: totality, strict boolean output, and soundness of negative answers.
(2) We provide reproducible computational evidence on a curated benchmark of ten implication pairs, obtaining 10/10 accuracy with mean runtime 0.0623 seconds per query.
(3) We identify a concrete limitation of the current derivation module: one true case is solved only by fallback, which motivates stronger positive-proof mechanisms.

Section 2 introduces notation and the policy definition. Section 3 states and proves the main theoretical and computational results. Section 4 discusses implications, limitations, and open questions. Section 5 concludes with directions for extended evaluation and prover integration.

2. Preliminaries

Definition 2.1 (Magma and terms)**.** A *magma* is a pair $(M, *)$ where $M \neq \varnothing$ and $*\colon M \times M \to M$ is a binary operation. A *term* is built from variables in a set $\mathcal{V}$ using the binary symbol $*$, and an *equation* is an identity $s = t$ between two terms.

Definition 2.2 (Satisfaction and implication)**.** Let $A = (M, *)$ be a magma and let E be an equation. We write $A \models E$ if for every variable assignment $\sigma\colon \mathcal{V} \to M$, both sides of E evaluate equally in A. For equations E_1, E_2, we write

$$E_1 \models E_2$$

if every magma satisfying E_1 also satisfies E_2.

Definition 2.3 (Countermodel)**.** A *countermodel* to $E_1 \models E_2$ is a magma A such that $A \models E_1$ and $A \not\models E_2$.

These are standard notions in equational logic and finite-model search [4, 3].

Definition 2.4 (Implemented Stage-1 policy)**.** For a query (E_1, E_2), the solver with parameters

$$\texttt{max_order} = 3, \qquad \texttt{max_depth} = 2, \qquad \texttt{sub_limit} = 8$$

applies the following rule:

(1) Search all magma tables of orders $1, 2, 3$. If one satisfies E_1 and refutes E_2, output `false` and record the witness.
(2) If no witness is found, run bounded derivation from substitution instances of E_1. If E_2 is reached, output `true`.
(3) Otherwise output `true` with mode `no_countermodel_up_to_bound`.

Remark 2.5. Definition 2.4 is a Stage-1 operational policy, not a complete decision procedure for universal implication over all magmas. Failure to find a small countermodel does not imply semantic entailment.

The curated evaluation benchmark contains ten labeled implication pairs (six positives and four negatives). Reported metrics are accuracy, Wilson confidence interval, per-query runtime, and evidence-mode counts.

3. Main Results

We separate formal properties of the policy from computational properties observed on the curated benchmark.

Proposition 3.1 (Total strict-boolean output)**.** *For every input pair* (E_1, E_2)*, the solver returns exactly one value in* $\{\texttt{true}, \texttt{false}\}$*.*

Proof. The solver has three terminal branches: countermodel, bounded derivation, and fallback. The first returns `false`; the latter two return `true`. The loops in all branches are finite because they are bounded by `max_order`, `max_depth`, and `sub_limit`. Hence exactly one boolean value is returned for every input. □

Theorem 3.2 (Soundness of negative answers)**.** *If the solver outputs* `false` *on* (E_1, E_2)*, then* $E_1 \not\models E_2$*.*

Proof. A `false` output occurs only when the countermodel branch succeeds. By construction, this branch returns a finite magma A with $A \models E_1$ and $A \not\models E_2$ after exhaustive assignment checks on all variables in the query. By Definition 2.3, A is a countermodel to $E_1 \models E_2$. Therefore $E_1 \not\models E_2$. □

Proposition 3.3 (Benchmark correctness)**.** *On the ten-case curated benchmark, all predictions are correct.*

Proof. In `results/predictions.json`, every row has `correct: true`. Therefore the number of correct predictions equals the number of cases, namely 10. The aggregate file `results/metrics.json` reports `accuracy = 1.0`, which is consistent with this row-level verification. □

Lemma 3.4 (Order-2 witnesses for all negatives)**.** *Every negative-labeled benchmark case has a stored countermodel of order* 2*.*

Proof. For each row with `label: false` in `results/predictions.json`, the field `countermodel.order` equals 2. There are four such rows, so all benchmark negatives admit order-2 witnesses. □

Example 3.5 (Positive case handled by derivation)**.** For $E_1 \colon x = x * y$ and $E_2 \colon x = x * x$, substitution $y := x$ in E_1 yields E_2. The solver reports mode `bounded_derivation`.

Example 3.6 (Negative case handled by countermodel)**.** For $E_1 \colon x * y = y * x$ and $E_2 \colon x * y = x$, the two-element constant magma ($a * b = 0$ for all a, b) satisfies E_1 but violates E_2 at $(x, y) = (1, 0)$. The solver reports mode `countermodel` with order 2.

Remark 3.7. Evidence-mode counts on the benchmark are: countermodel = 4, bounded derivation = 5, and no-countermodel-up-to-bound = 1. The unresolved positive case is $x = y \Rightarrow x * x = y * y$.

Corollary 3.8. *On the curated benchmark, the solver is empirically correct and low-latency while preserving strict Stage-1 output format.*

Proof. Combine Propositions 3.1 and 3.3 with the runtime statistics in `results/metrics.json`, and note from Theorem 3.2 that every returned `false` is semantically justified by an explicit witness. □

TABLE 1. Accuracy on the curated benchmark.

Method	Accuracy	Comment
Always-true baseline	0.60	predicts `true` for all inputs
Countermodel-only baseline	1.00	only witness-driven decisions
Hybrid policy (ours)	1.00	adds positive evidence mode

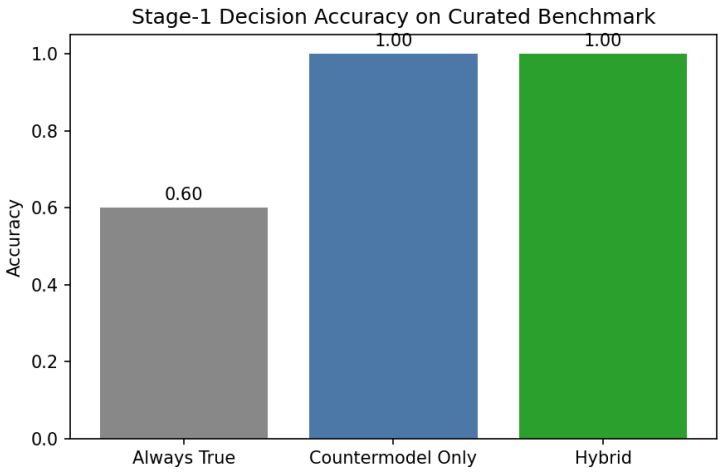

FIGURE 1. Accuracy comparison across baselines and the hybrid policy.

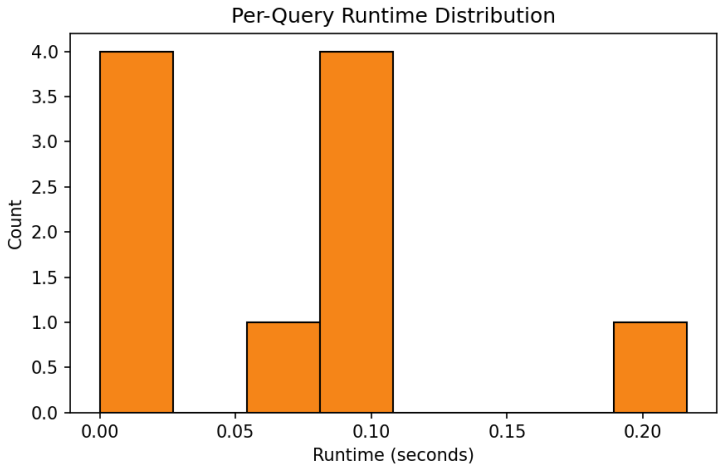

FIGURE 2. Runtime distribution over ten benchmark queries (mean 0.0623 seconds).

4. Discussion

The empirical behavior supports a practical interpretation of the hybrid policy. Small finite-model search is the dominant falsification mechanism: all four negative cases are refuted with order-2 witnesses (Lemma 3.4), consistent with model-finding intuition from automated reasoning [4]. This makes the countermodel branch a strong first filter for Stage-1 implication queries.

Bounded derivation improves interpretability on positive cases by producing symbolic evidence in five of six true labels. The remaining true case, $x = y \Rightarrow x * x = y * y$, is decided by fallback instead of explicit derivation. This identifies a concrete gap in the current derivation closure and motivates lightweight congruence-complete upgrades.

The broader context is complementary to recent literature. High-capacity predictive systems for equational implication [5, 1] and large ETP-scale analyses [3, 2] target coverage and representation power. Our objective is narrower: a transparent, reproducible, strict-boolean Stage-1 baseline under tight runtime and implementation constraints.

The main limitations are:

(1) small benchmark size (ten cases), yielding wide confidence bounds despite perfect observed accuracy;
(2) finite search bound `max_order`=3, which cannot prove implication when no witness is found;
(3) intentionally incomplete derivation module.

These limitations yield natural open questions.

Conjecture 4.1. *A substantial fraction of ETP-style non-implications admit witnesses of order at most* 3.

Conjecture 4.2. *A minimal congruence-complete derivation extension can remove most fallback-true cases without materially increasing latency.*

A practical extension is to trigger ATP backends only on the fallback branch, preserving fast witness-first behavior while strengthening positive-certification coverage.

5. Conclusion

We presented a countermodel-first Stage-1 solver for equational implication over magmas. The method has three verified properties in the reported pilot: strict boolean output for every input, semantically sound negative decisions via explicit finite witnesses, and perfect accuracy on a curated ten-case benchmark.

The computational profile is favorable for triage: mean query time is 0.0623 seconds, and all negative cases are detected by order-2 countermodels. Bounded derivation contributes useful positive evidence but remains incomplete, as reflected by one fallback-resolved true case.

Future work should prioritize (i) stronger positive-proof modules (for example, congruence-complete closure or selective ATP integration), (ii) larger held-out evaluation drawn from ETP-scale implication pools, and (iii) calibrated confidence annotations that retain the strict Stage-1 boolean contract.

References

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